



PPS - 41

June 6, 2026 – Paestum, Salerno

Introduction to OpenFOAM® Computational Library and Design of Profile Extrusion Dies

P3 - Hands-On Profile Extrusion Dies

J. Miguel Nóbrega

mnobrega@dep.uminho.pt

Mohammadreza Aali

mohammadreza.aali@jku.at

Outline

10:00 – 11:15	Introduction to OpenFOAM and Simulation Fundamentals (P1)
11:30 – 13:00	Geometry, Mesh Generation, and Case Setup in OpenFOAM (P2)
13:00 – 15:00	Lunch break
15:00 – 17:00	Hands-On Profile Extrusion Dies (P3)



PPS - 41



IPC

JYU IPPD Institute of
Polymer Processing and
Digital Transformation

Session Contents

- Presentation
- Geometry generation (FreeCAD)
- Mesh generation
- Solver *hpc4PEFoam*
- Bird Carreau + Arrhenius Constitutive model
- Objective Function and convergence
- Run and test



PPS - 41



CRheo@IPC

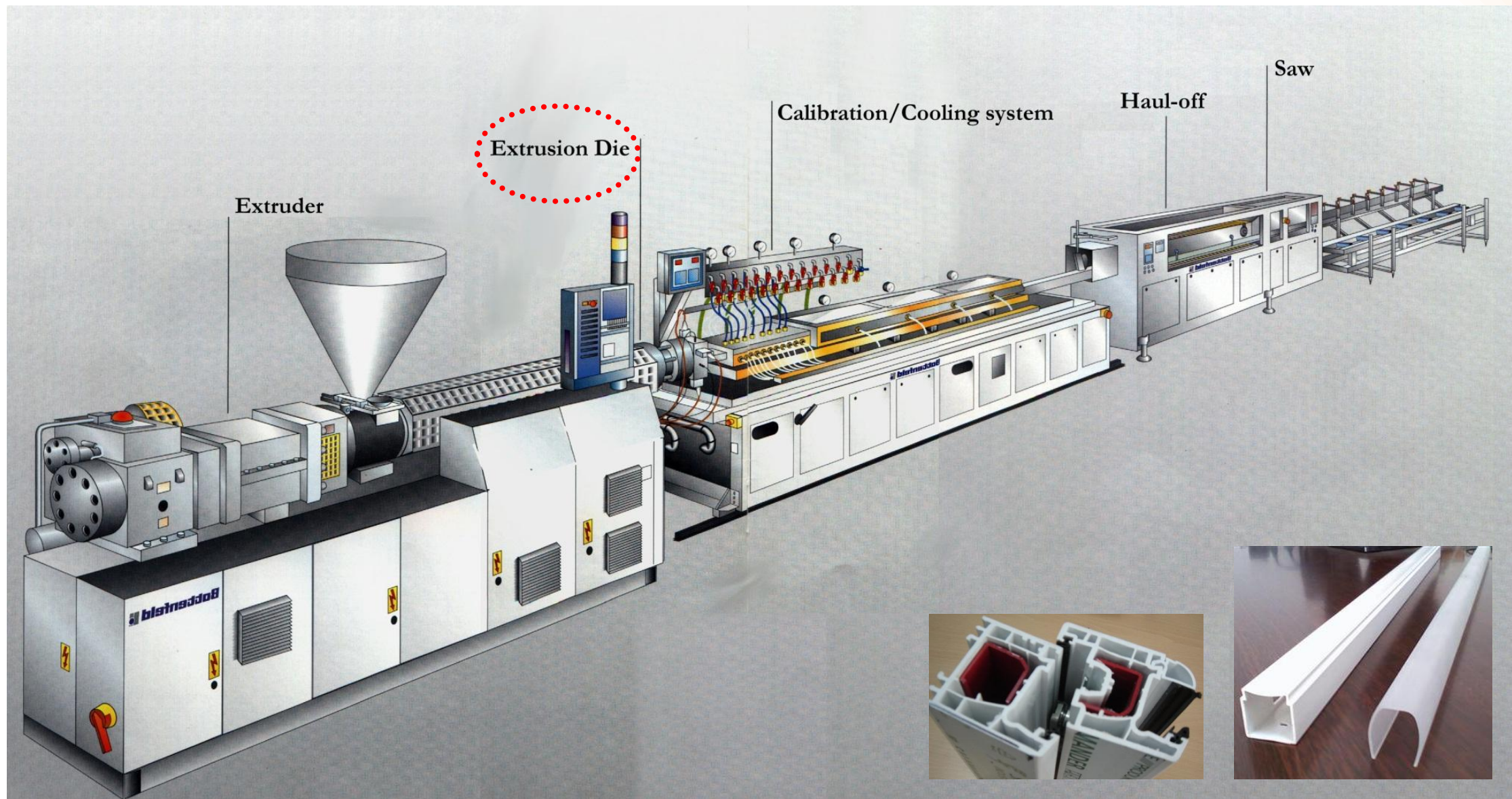


IPC

JYU

IPPD Institute of
Polymer Processing and
Digital Transformation

Introduction



PPS - 41



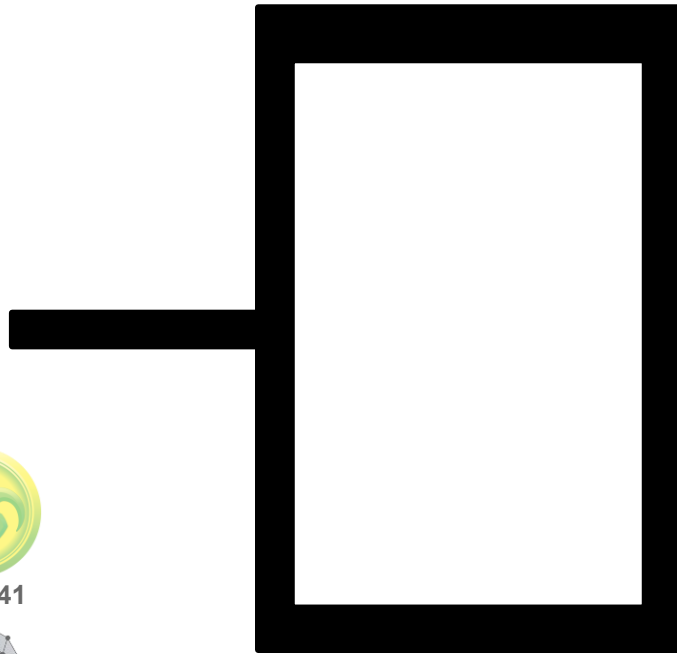
IPC

JYU IPPD Institute of
Polymer Processing and
Digital Transformation

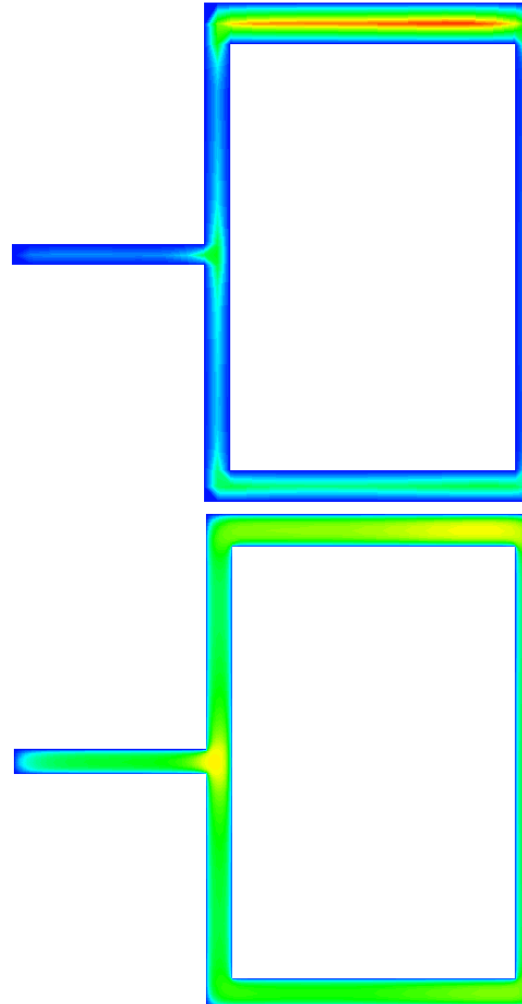
Introduction to OpenFOAM® Computational Library and Design of Profile Extrusion Dies

Introduction – Flow Balance

Die land cross-section



Velocity contours



Extrusion run



Unbalanced



Balanced



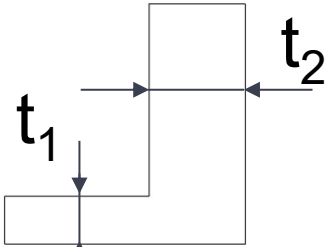
OS Carneiro, JM Nóbrega (Editors) , Design of Extrusion Forming Tools, Smithers Rapra Technology, ISBN 9781847355171, 2012.

JMPPD Institute of Polymer Processing and Digital Transformation, FT Pinho, PJ Oliveira International Polymer Processing, 2004

Introduction to OpenFOAM® Computational Library and Design of Profile Extrusion Dies

Introduction – Optimisation Strategies

Required Profile

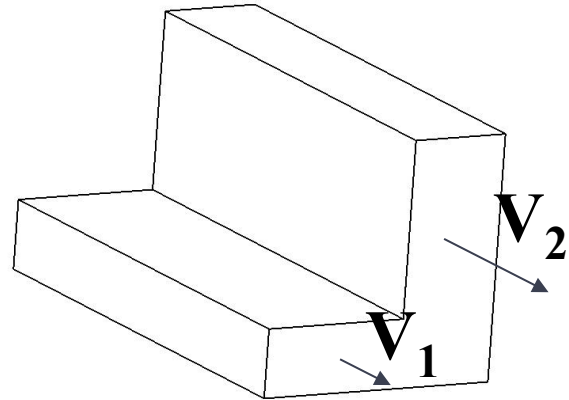
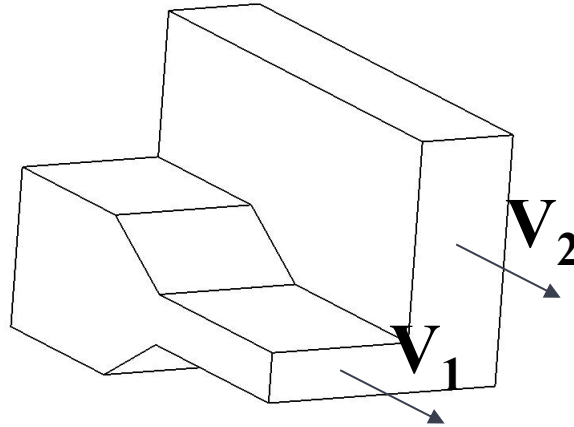


Optimisation Strategy

ST1
(length)

ST2
(thickness)

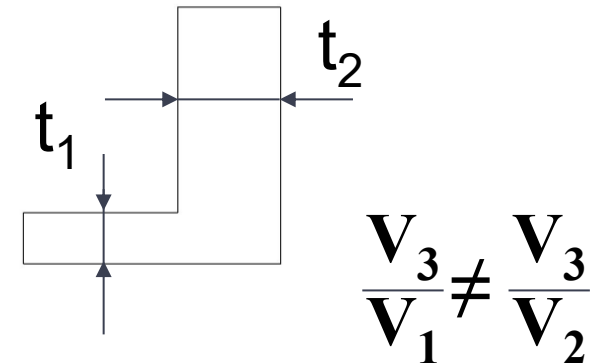
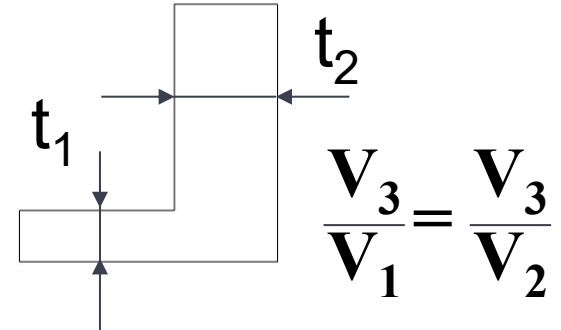
Die Flow Channel



Haul-off Speed

V_3

Final Profile



Implementation in OpenFOAM

Methodology Employed

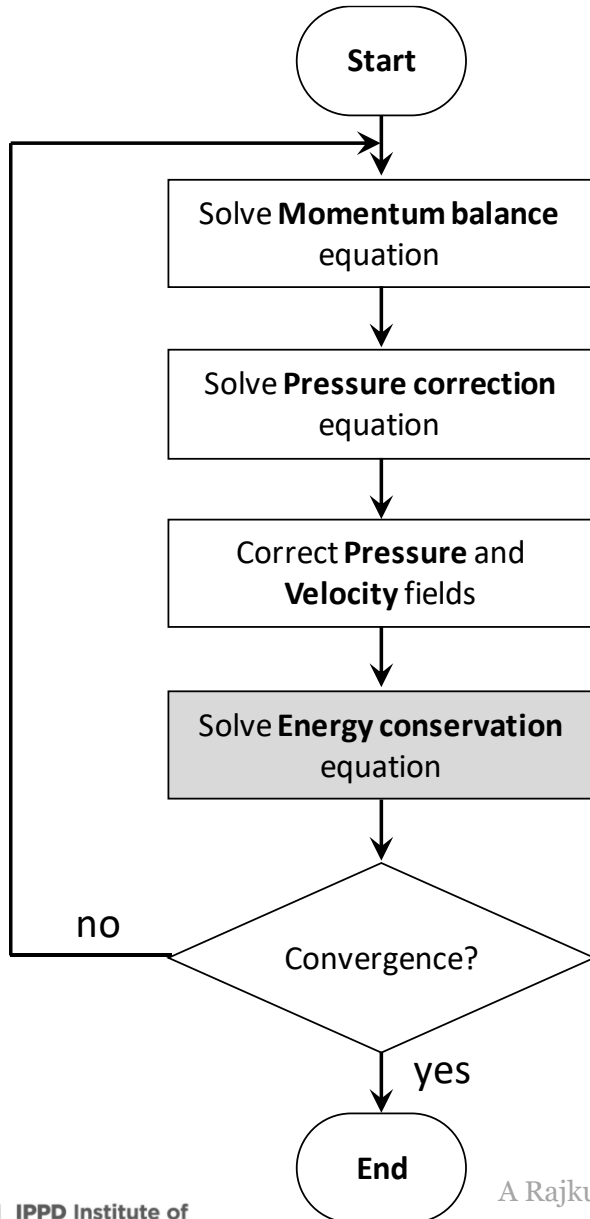
Energy conservation equation (Steady State)

$$\nabla \cdot (UT) - \nabla \cdot \left(\frac{k}{\rho c_p} \nabla T \right) = \frac{1}{\rho c_p} \tau : \nabla U$$

OpenFOAM Code

Solve

```
(  
    fvm :: div (phi , T)  
    - fvm :: laplacian (DT , T)  
    == (1 / c_p) * (tau && grad U)  
)
```



A Rajkumar et al., An Open-source framework for the CAD of Complex Profile Extrusion Dies, International Polymer Processing, 33, 2018

Introduction to OpenFOAM® Computational Library and Design of Profile Extrusion Dies



PPS - 41



IPC

JYU

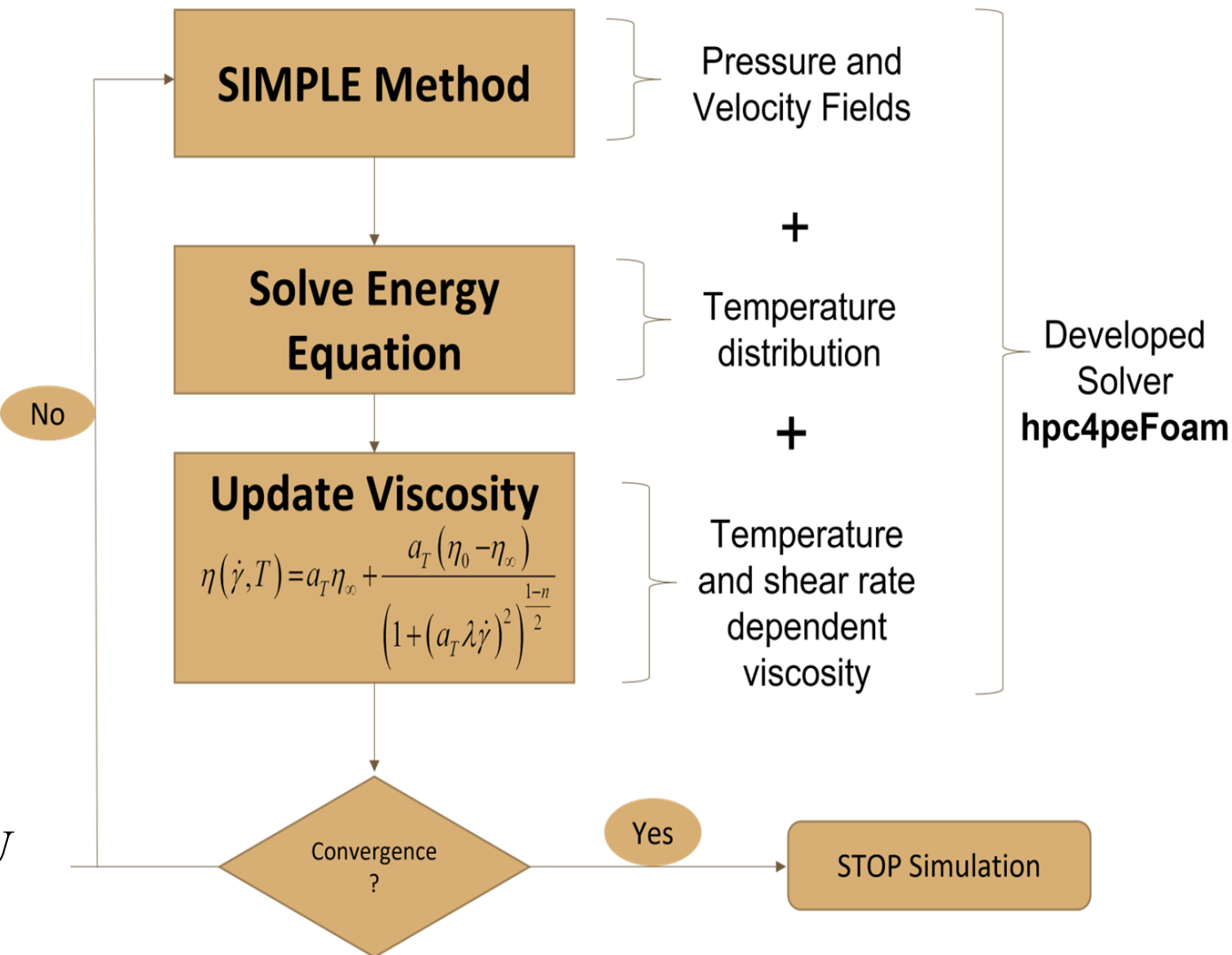
IPPD Institute of
Polymer Processing and
Digital Transformation



Open  FOAM

Energy conservation equation
(Steady State)

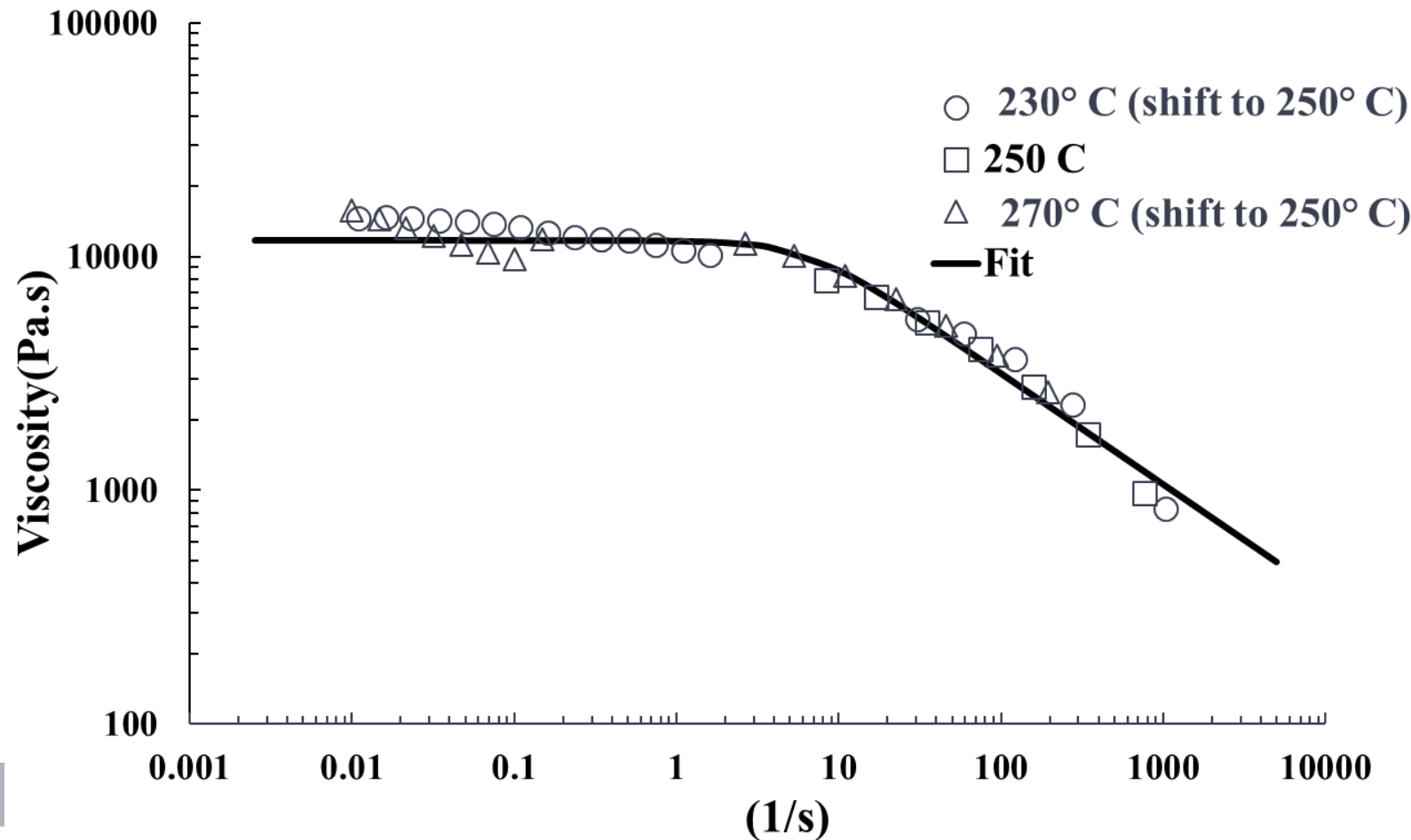
$$\nabla \cdot (UT) - \nabla \cdot \left(\frac{k}{\rho c_p} \nabla T \right) = \frac{1}{\rho c_p} \boldsymbol{\tau} : \nabla \mathbf{U}$$



Implementation in OpenFOAM

Material characterization

Polycarbonate



Bird Carreau + Arrhenius Law

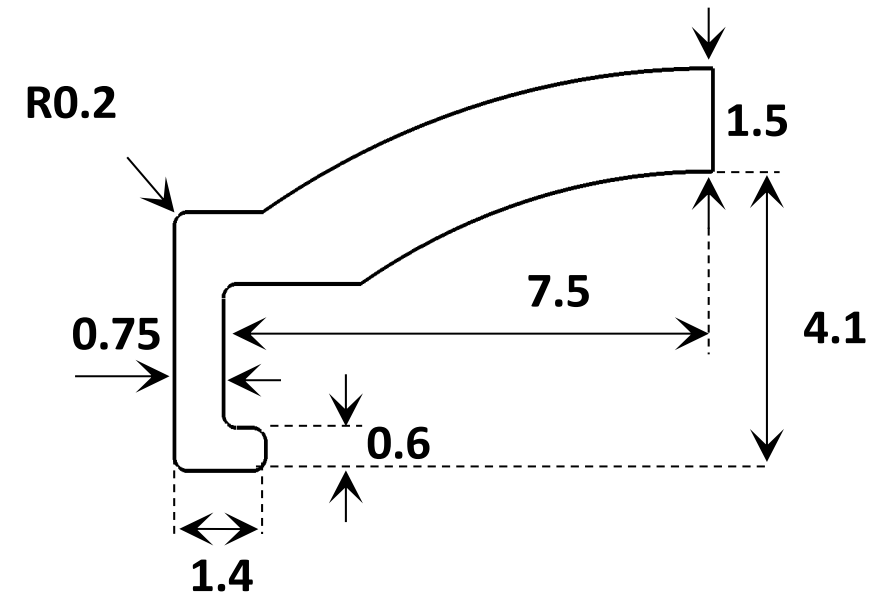
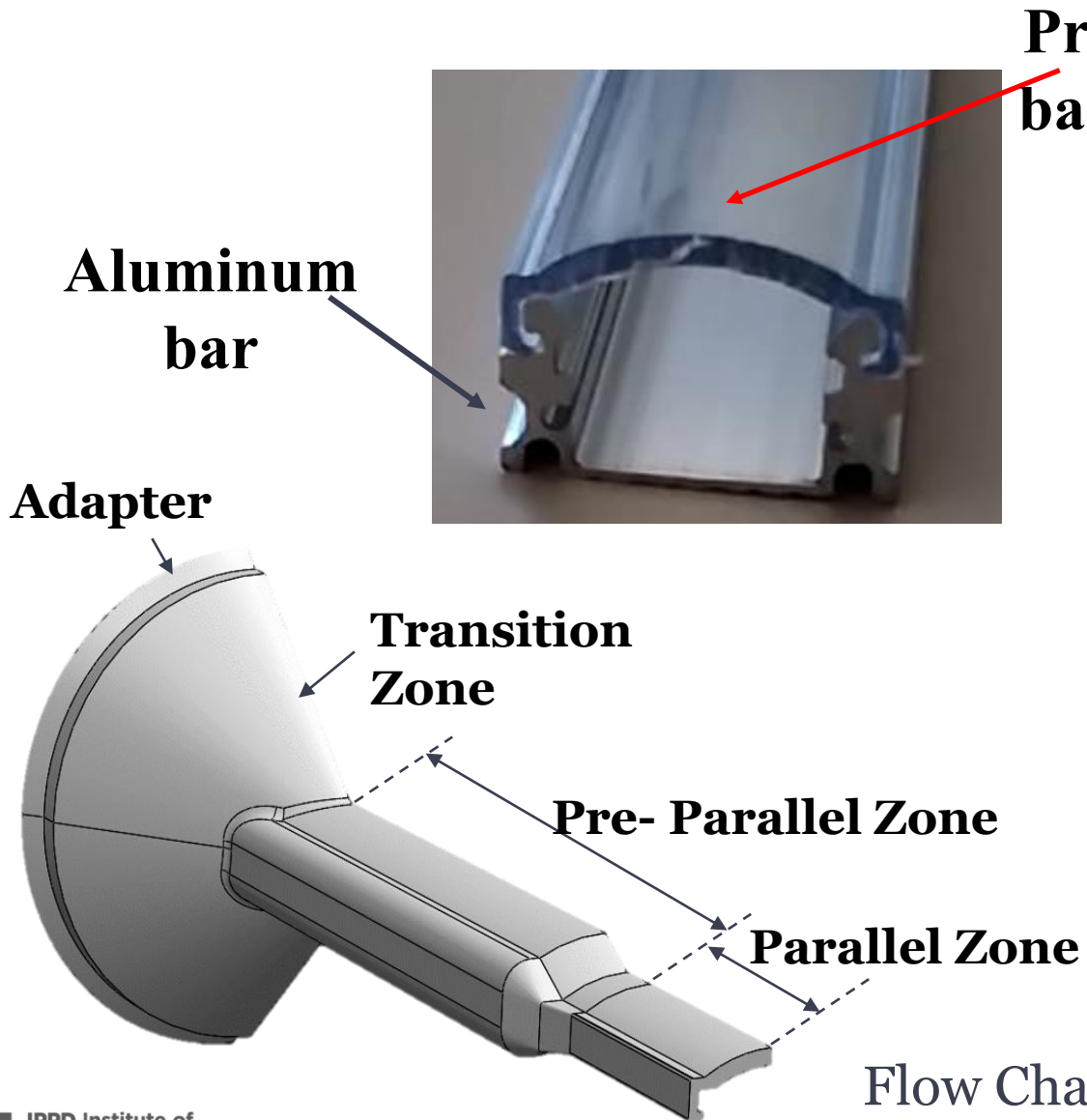
$$\eta(\dot{\gamma}, T) = a_T \eta_\infty + \frac{a_T (\eta_0 - \eta_\infty)}{\left(1 + (a_T \lambda \dot{\gamma})^2\right)^{\frac{1-n}{2}}}$$

Shift Factor

$$a_T = \exp\left(\frac{E}{R} \left(\frac{1}{T} - \frac{1}{T_0}\right)\right)$$



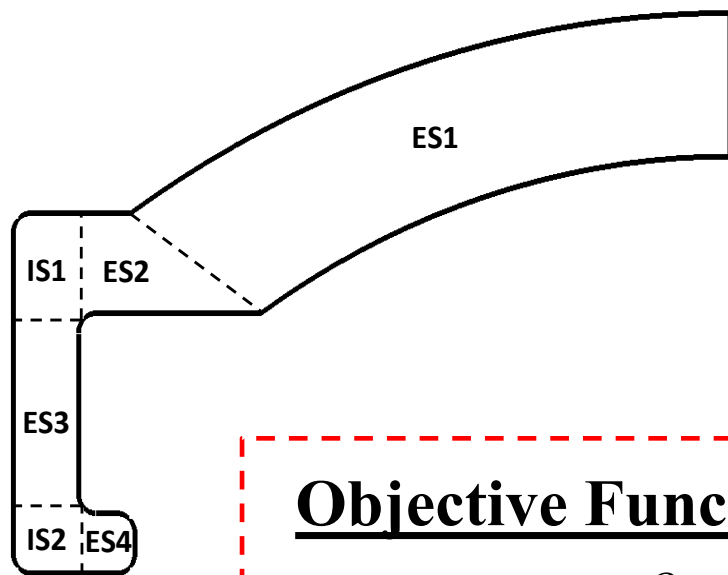
Implementation – Case Study



Profile Cross Section (*dimensions in mm*)

Flow Channel Geometry

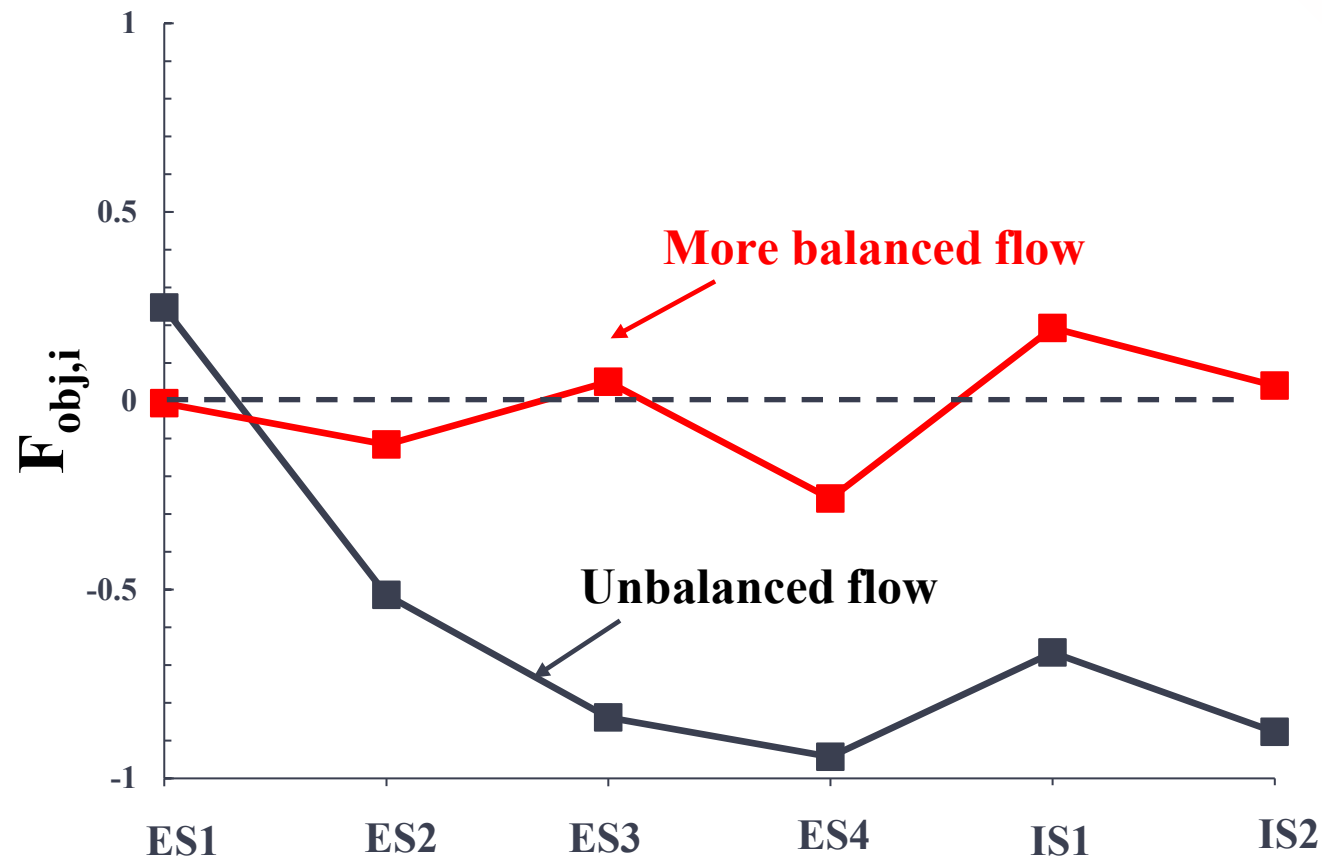
Implementation – Case Study



Objective Function

$$F_{obj,i} = \frac{\frac{Q_i}{Q_{target}} - 1}{\max\left(\frac{Q_i}{Q_{target}}, 1\right)}$$

$$F_{obj} = \frac{\sum_{ES+IS} \|F_{obj,i}\| A_{target,i}}{A_{target,tot}}$$

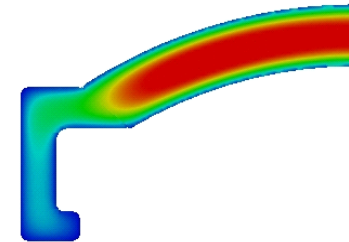
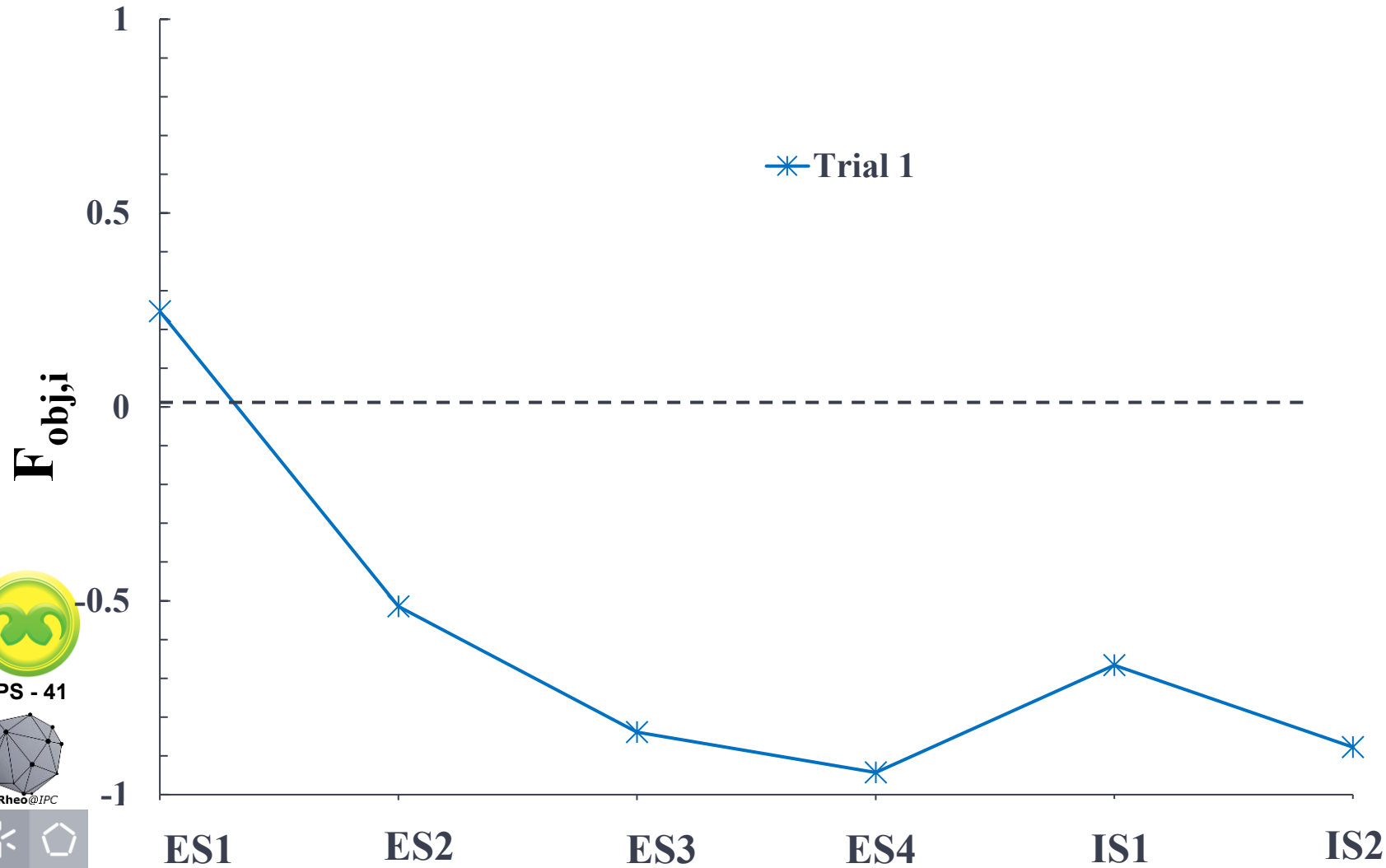


PPS - 41

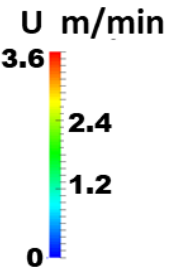


IPPD Institute of
Polymer Processing and
Digital Transformation

Implementation – Case Study

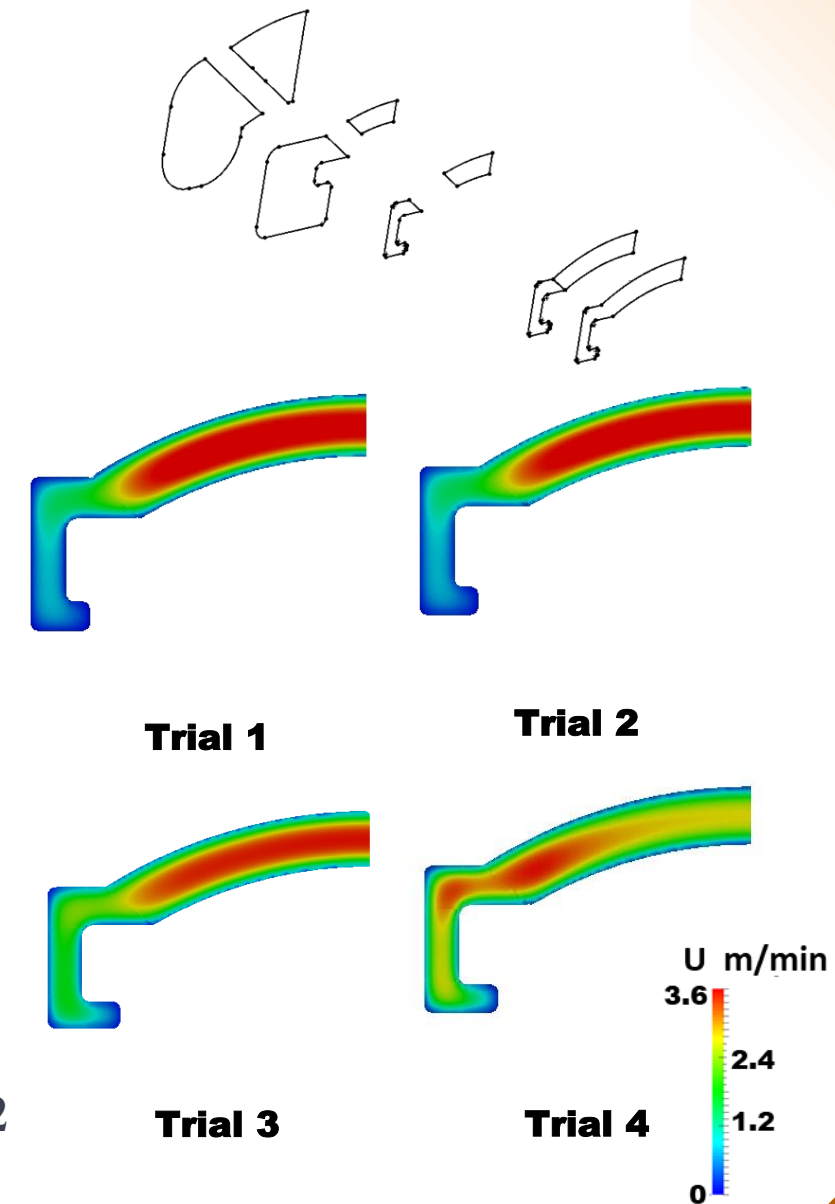
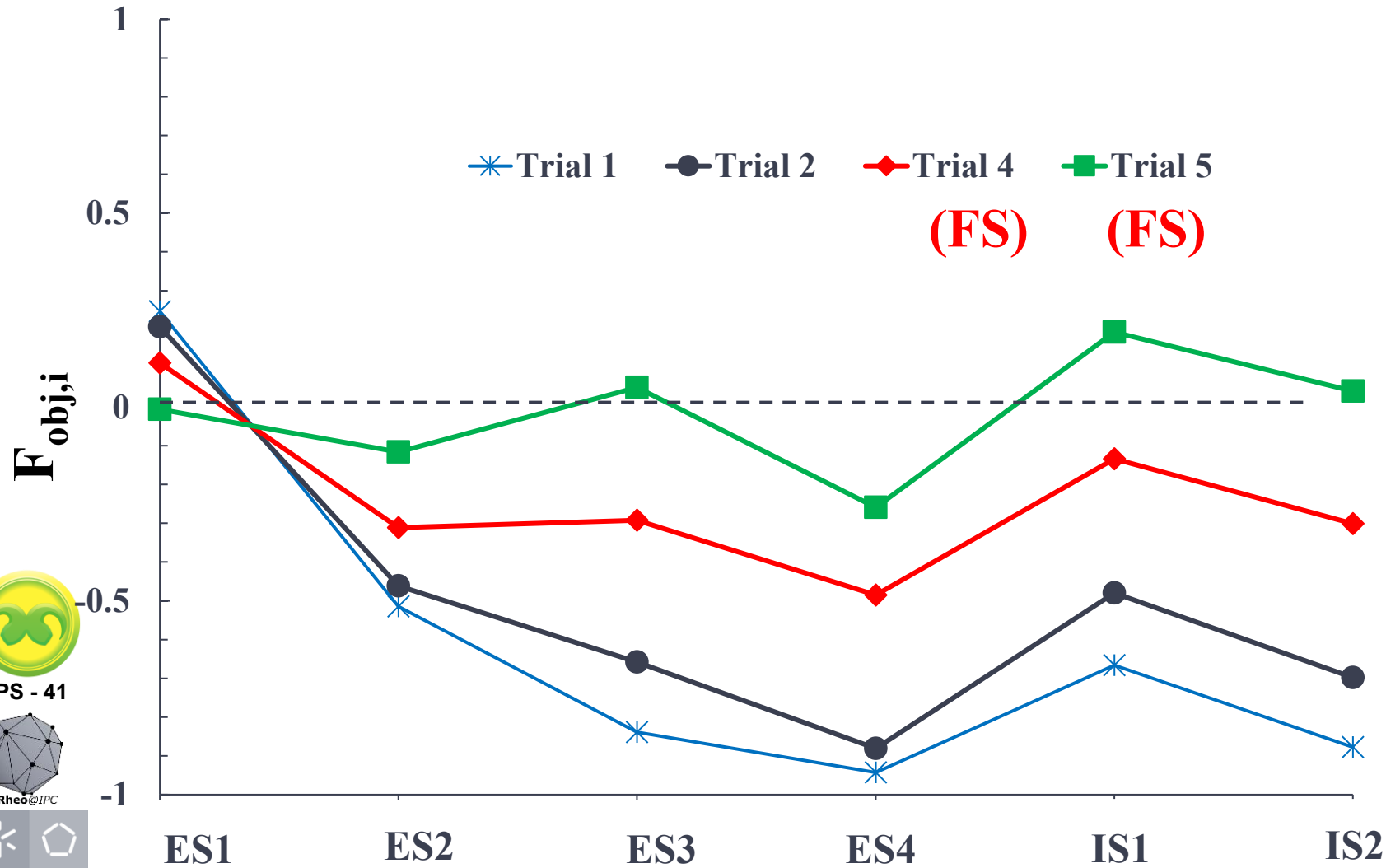


Trial 1



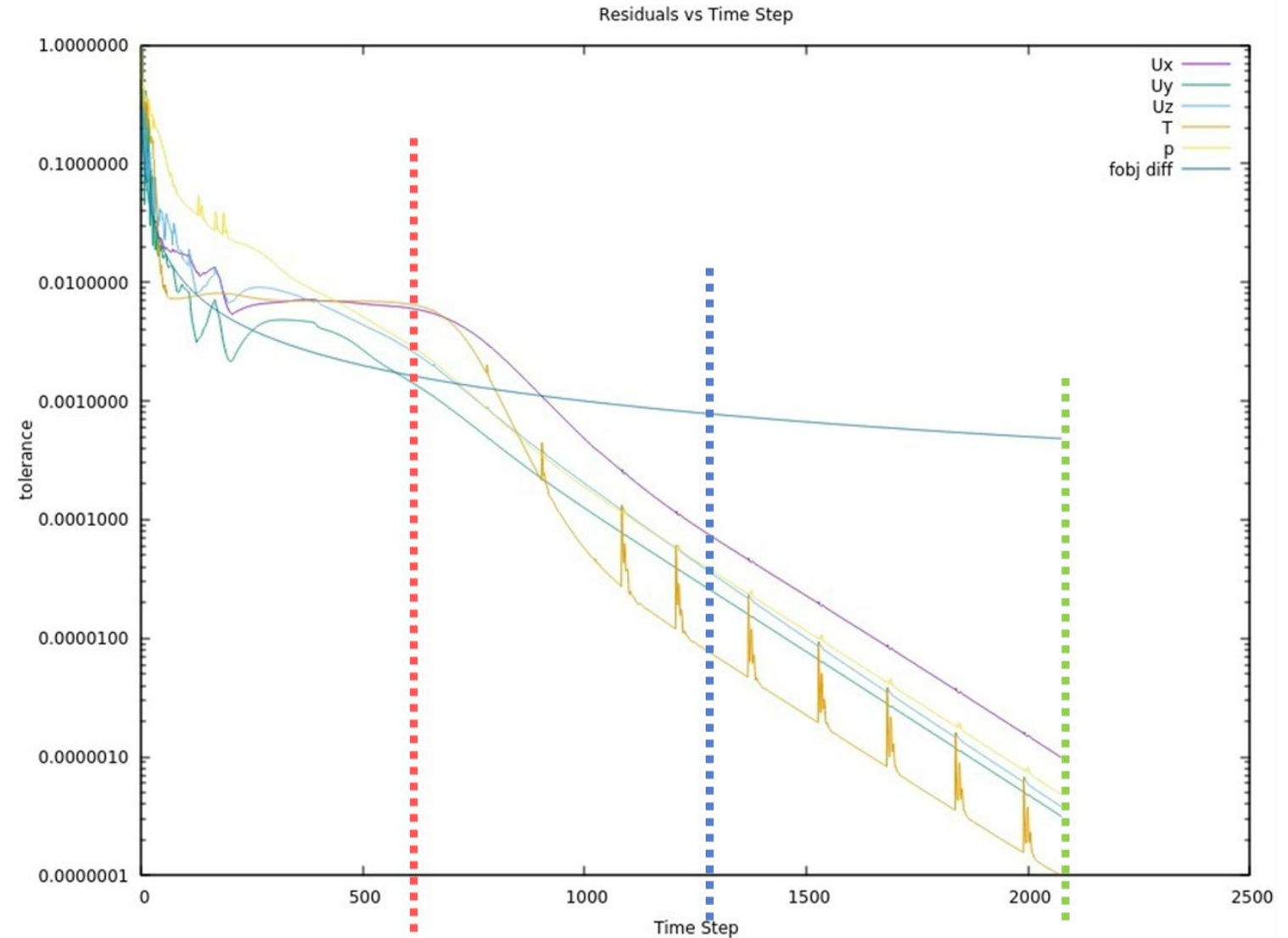
Implementation – Case Study

Flow Separator (FS)



Convergence Criteria

Based of the F_{obj} variation



Simulation stops when
Objective function stabilizes
(tol < 1%)

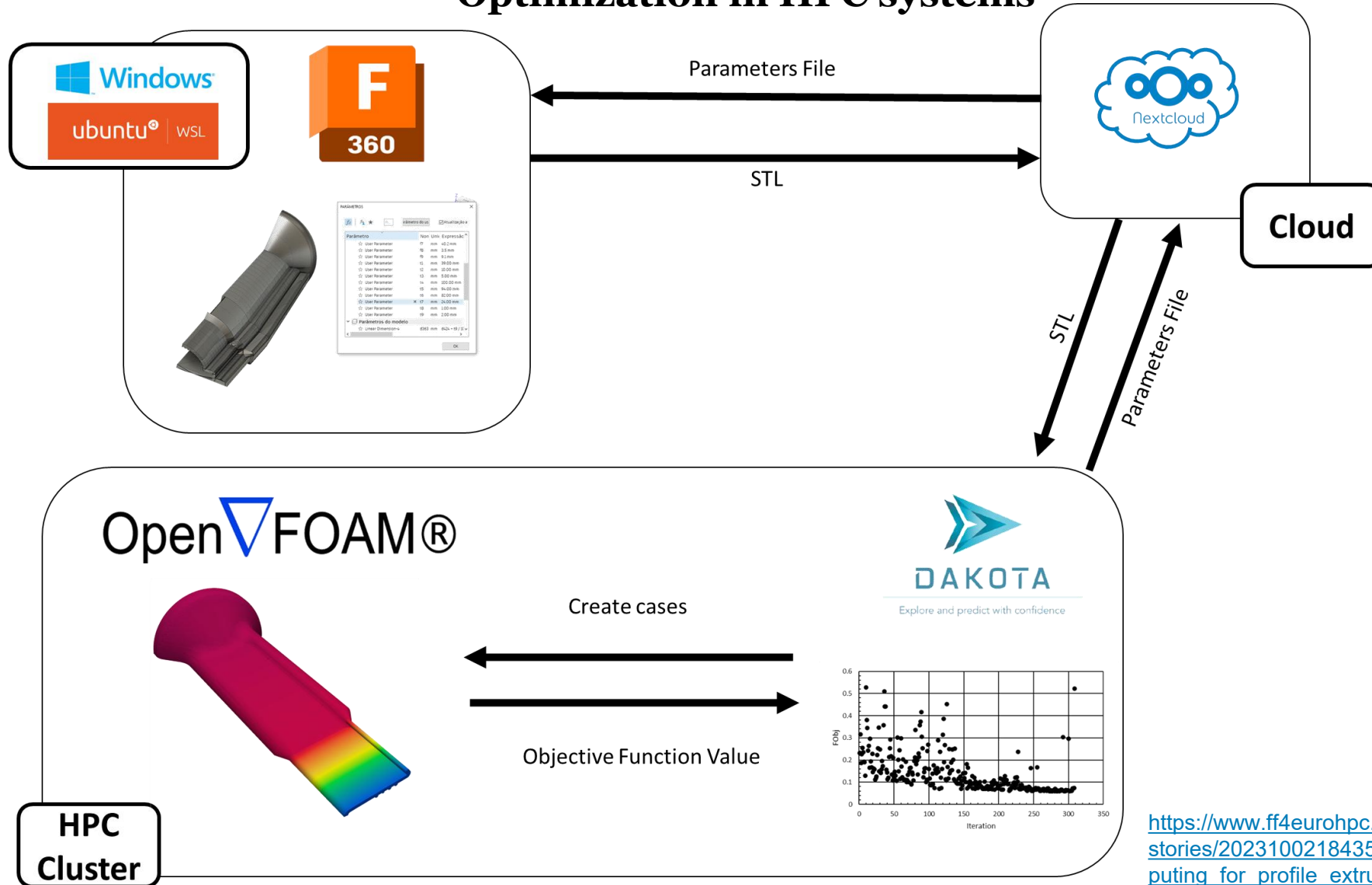
Simulation stops based on
Residuals (U, p and T converge
when reach 10^{-4})

Simulation stops based on
Residuals (U, p and T
converge when reach 10^{-8})



hpc4PE – Calculation Framework

Optimization in HPC systems



Vidal, J. P. O., et al. (2024). Enhancing extrusion die design efficiency through high-performance computing based optimization. Meccanica, 1-12.

Introduction to OpenFOAM® Computational Library and Design of Profile Extrusion Dies

https://www.ff4eurohpc.eu/en/success-stories/2023100218435592/high_performance_computing_for_profile_extrusion

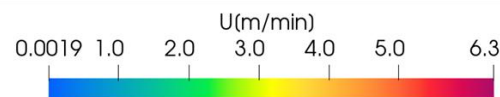
hpc4PE Case studies



Optimization in HPC systems

		F _{obj} Improvement	Velocity Distribution		Cores	Trials	Time (Hours)	Cost (€)			
ID	Material		Initial	Final							
P3	HDPE	2.3			512	409	23	145			
			• Reduction of 30-40% on the product time to market • Reduction at least in 40% of the raw materials spent in experimental trials • Estimated savings of 23 % in tools development					77			
								115			

- Reduction of 30-40% on the product time to market
- Reduction at least in 40% of the raw materials spent in experimental trials
- Estimated savings of 23 % in tools development



https://www.ff4eurohpc.eu/en/success-stories/2023100218435592/high_performance_computing_for_profile_extrusion

Vidal, J. P. O., et al. (2024). Enhancing extrusion die design efficiency through high-performance computing based optimization. Meccanica, 1-12.

Introduction to OpenFOAM® Computational Library and Design of Profile Extrusion Dies



PPS - 41



IPC

JYU

IPPD Institute of
Polymer Processing and
Digital Transformation

Acknowledgements

This work was funded by National Funds through FCT - Portuguese Foundation for Science and Technology, Reference UID/CTM/50025/2019 and UIDB/04436/2020. The experiment **“HPC4PE - High Performance Computing for Profile Extrusion”** has received funding from the **European High-Performance Computing Joint Undertaking (JU)** through the **FF4EuroHPC project under grant agreement No 951745**. The JU receives support from the European Union’s Horizon 2020 research and innovation programme and Germany, Italy, Slovenia, France, Spain. The authors also acknowledge the support of the computational cluster MACC.

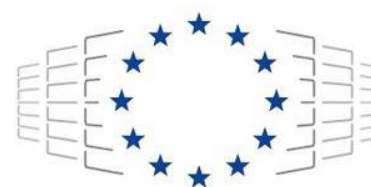


PPS - 41



IPC

JYU IPPD Institute of
Polymer Processing and
Digital Transformation



EuroHPC
Joint Undertaking